

Gemini3ProPreview

Q: I am a student who is a first-time learner of Python and its packages. Please provide an at-a-glance summary of how pandas has been used in the attached marimo notebooks (Wk02_Marimo_ZScore.py, Wk03_InteractivePlots_Plotly.py, Wk04_DataPreparation.py) to help me understand its usage better.

Here is an at-a-glance summary of how **pandas** is used across your notebooks. Since you are a first-time learner, I have grouped these commands by **what they accomplish** rather than by file.

Think of pandas as "Excel for Python"—it is used to load, view, clean, and calculate data using rows and columns.

1. Loading and Saving Data

How you get data into Python and how you save your results.

Command	Used In	Description
<code>pd.read_csv('filename.csv')</code>	Wk03, Wk04	Loads a CSV file into a pandas DataFrame (a table).
<code>df.to_csv('filename.csv')</code>	Wk02, Wk04	Saves your DataFrame back to a CSV file.
<code>yfinance</code> output	Wk02	<i>Note:</i> The <code>yfinance</code> library returns data (like <code>stock.balance_sheet</code>) directly as pandas DataFrames or Series.

2. Selecting and Viewing Data

How you pick specific parts of your table to look at or use.

Command	Used In	Description
<code>df</code>	All	Typing just the variable name displays the table (in Marimo/Jupyter).

Command	Used In	Description
<code>.iloc[:, 0]</code>	Wk02	Integer Location. Selects all rows (<code>:</code>) and the <i>first</i> column (<code>0</code>). Used to get the most recent financial data.
<code>.get('Key')</code>	Wk02	Safely retrieves a value from a Series (a single column). If the key is missing, it returns <code>None</code> instead of crashing.
<code>df['Column_Name']</code>	All	Selects a specific column (e.g., <code>df['Total_Assets']</code>).
<code>df[['Col1', 'Col2']]</code>	Wk04	Selects a subset of specific columns to view.

3. Filtering Data

How you isolate specific rows based on conditions (like "Show me only Apple" or "Show me companies with no debt").

Command	Used In	Description
<code>df[df['Col'] == 'Value']</code>	Wk03	Boolean Indexing. Keeps rows where the column equals a specific value (e.g., <code>df[df['Ticker'] == 'MSFT']</code>).
<code>df[df['Col'].isna()]</code>	Wk03	Filters to show only rows where data is missing (NaN).
<code>df.dropna(subset=[...])</code>	Wk04	Removes rows where specific columns have missing data.
<code>.unique()</code>	Wk03	Returns a list of all unique values in a column (e.g., all unique sectors).

4. Data Cleaning

How you fix "messy" data (missing values or wrong types).

Command	Used In	Description
<code>.fillna(0)</code>	Wk03	Replaces "Not a Number" (NaN) with 0. Used for companies that didn't report debt.

Command	Used In	Description
<code>.copy()</code>	Wk03, Wk04	Creates a standalone copy of the data. Important so you don't accidentally modify the original source data.

5. Calculations (Vectorization)

This is the most powerful part of pandas. Instead of writing a loop to calculate "Assets - Liabilities" for 500 companies one by one, pandas does it all at once.

Command	Used In	Description
<code>df['New_Col'] = df['A'] / df['B']</code>	Wk02, Wk04	Vector Math. Divides Column A by Column B for every row instantly and saves it as 'New_Col'.
<code>df.sort_values(['Col1', 'Col2'])</code>	Wk04	Sorts the data (e.g., by Ticker, then by Year) so that calculations order matters.
<code>.groupby('Ticker')</code>	Wk04	Grouping. Tells pandas to treat each company separately.
<code>.shift(1)</code>	Wk04	Lagging. Moves data down by 1 row. Combined with <code>groupby</code> , this grabs the <i>previous year's</i> data for the <i>same</i> company.
<code>.apply(lambda x: ...)</code>	Wk02, Wk04	Applies a custom function to every single row. Used to categorize Z-Scores into "Safe", "Grey", or "Distress" zones.

6. Analysis & Summarization

How you create summary tables for the final report.

Command	Used In	Description
<code>pd.crosstab(...)</code>	Wk04	Creates a "Contingency Table" (like a Pivot Table in Excel) to count how many companies fall into specific categories (e.g., High Cost of Debt vs. Low Risk).

Summary for the Beginner

1. **Wk02** used pandas mostly to **fetch** single values (`.iloc` , `.get`) to calculate a Z-Score for one company.
2. **Wk03** used pandas to **clean** (`.fillna`) and **filter** (`df[...]`) a large dataset for visualization.
3. **Wk04** used pandas for **time-series logic**. It used `groupby` and `shift` to look at past data (Lagged Z-Score) and compare it to current data.